

Vendor Performance Analysis Report

Executive Summary

This report presents an analytical study of vendor performance using a custom-built data pipeline. Raw datasets were ingested into a database, transformed and cleaned, merged using SQL operations, and analyzed to identify business patterns and performance indicators.

Methodology

The workflow involved data ingestion from large files, database storage using SQLite, data cleaning, feature engineering, merging multiple datasets, and exploratory analysis. Key calculated metrics included Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio.

Key Findings

Products with negative gross profit indicate potential loss-making transactions. Certain inventory items were purchased but not sold, suggesting slow-moving stock. Strong relationships between purchase quantity and sales quantity indicate effective inventory movement.

Business Insights

Brands with lower sales but higher profit margins were identified as potential candidates for targeted promotion and pricing strategies. Correlation analysis suggested that pricing variations alone may not strongly influence profitability.

Recommendations

Review loss-making products, improve inventory optimization, promote high-margin products with weak sales performance, and investigate pricing patterns for stronger revenue generation.

Analysis Visualizations



